



Student Name

Teacher Name:

2018

**HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION**

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided at the back of this paper
- All necessary working should be shown in Questions 11-16
- Write your student name at the top of the relevant pages

**Total marks:
100**

Section I – 10 marks (pages 2 – 3)

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 4 – 9)

- Attempt Questions 11-16
- Allow about 2 hours and 45 minutes for this section

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for questions 1-10

1 What is the value of $9.8 \times 10^7 - 2.3 \times 10^4$?

- A. 7.5×10^3
- B. 7.5×10^4
- C. 9.7977×10^6
- D. 9.7977×10^7

2 What is the value of $\sum_{k=1}^4 (-1)^k k^2$?

- A. -30
- B. -10
- C. 10
- D. 30

3 Which of the following quadratic expressions is positive definite?

- A. $x^2 + 5x + 2$
- B. $x^2 + 5x + 4$
- C. $x^2 + 5x + 6$
- D. $x^2 + 5x + 8$

4 Which of the following trigonometric expressions is equivalent to $\tan\left(\frac{\pi}{2} - x\right)$?

- A. $\tan x$
- B. $\cot x$
- C. $-\tan x$
- D. $-\cot x$

- 5 What is the range of the function $f(x) = \sqrt{1-x^2}$?
- A. $0 < y < 1$
B. $0 \leq y \leq 1$
C. $-1 < y < 1$
D. $-1 \leq y \leq 1$
- 6 What is the value of x if $\log x^2 - \log 3x = \log 9$?
- A. 3
B. 18
C. 27
D. 81
- 7 What is the value of $\int_{-2}^2 |x| dx$?
- A. 0
B. 2
C. 4
D. 8
- 8 What are the amplitude and period of the function $f(x) = 2 - \sin 2x$?
- A. Amplitude 1, period π
B. Amplitude 1, period 2π
C. Amplitude 2, period π
D. Amplitude 2, period 2π
- 9 Which of the following is an expression for $\frac{d}{dx}(e^{2x} \tan x)$?
- A. $(1 + \tan x)^2 e^{2x}$
B. $2(1 + \tan^2 x) e^{2x}$
C. $2e^{2x} \tan x$
D. $e^{2x} \sec^2 x$
- 10 What is the number of real roots of the equation $x(x-2)\log_e x = 0$?
- A. 0
B. 1
C. 2
D. 3

Section II

90 marks

Attempt Questions 11-16

Allow about 2 hours and 45 minutes for this section

Answer the questions in writing booklets provided. Use a separate writing booklet for each question. In Questions 11-16 your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a separate writing booklet.	Marks
(a) Find in simplest exact form the value of $(3 - 2\sqrt{2})^2$.	2
(b) Solve the quadratic equation $2x^2 - 5x - 3 = 0$.	2
(c) Differentiate with respect to x	
(i) $\sin(1 - x)$.	1
(ii) $x \log_e x$.	1
(d) Find the equation of the tangent to the curve $y = \frac{x}{x-1}$ at the point $(2, 2)$ on the curve.	3
(e) Evaluate in simplest exact form $\int_1^2 \frac{x+1}{x} dx$.	3
(f) The region bounded by the curve $y = \frac{1}{2x+1}$ and the x axis between $x = 0$ and $x = 1$ is rotated about the x axis to form a solid. Find the volume of the solid of revolution in simplest exact form.	3

Question 12 (15 marks) Use a separate writing booklet**Marks**

(a) Find $\int \sec x (\cos x + \sec x) dx$. **2**

(b) Find the coordinates of the point of inflexion on the curve $y = x^3 - 3x^2 + 6x$. **3**

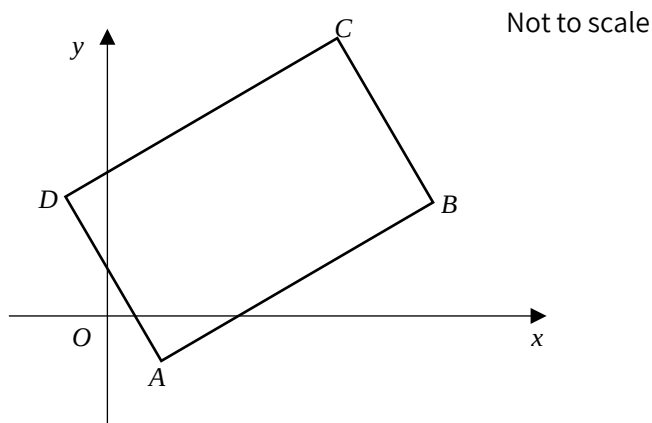
(c) A parabola has equation $8y = x^2 - 6x + 1$.

(i) Write the equation in the form $(x - h)^2 = 4a(y - k)$. **1**

(ii) Find the vertex of the parabola. **1**

(iii) Sketch the parabola showing vertex, directrix and focal point (label correctly and use a ruler). **3**

- (d) In the diagram below, $ABCD$ is a parallelogram. Side AB has equation $x - 2y - 3 = 0$, side BC has equation $2x + y - 16 = 0$ and vertex D has coordinates $(-1, 3)$.

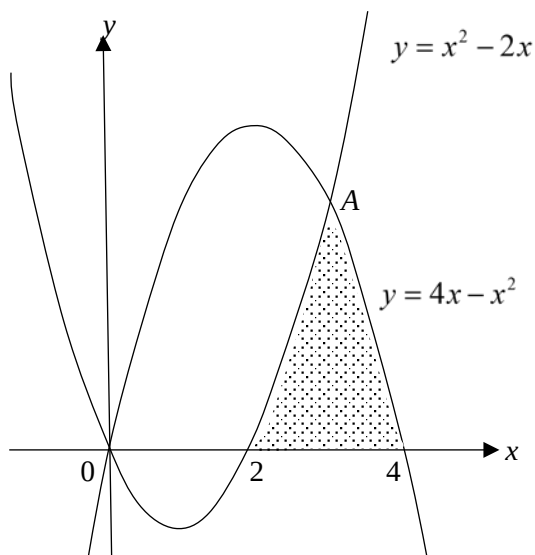


(i) Show that $ABCD$ is a rectangle. **2**

(ii) Find the coordinates of the vertex C . **3**

- (a) The gradient function of a curve $y = f(x)$ is given by $f'(x) = 4x^3 + x^2 + 7$. 3
The curve passes through the point $(4, 5)$. Find the equation of the curve.

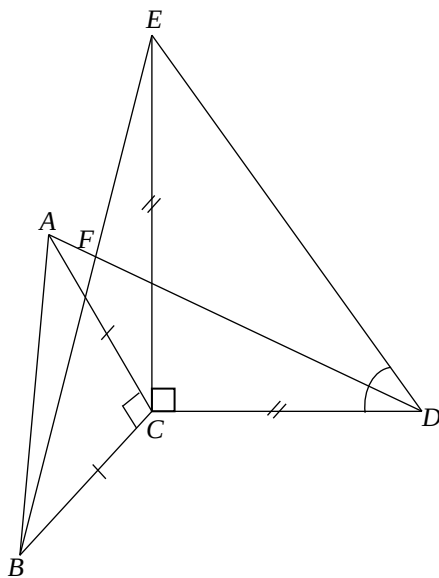
- (b) The diagram below shows the parabolas $y = 4x - x^2$ and $y = x^2 - 2x$. The graphs intersect at the origin O and the point A .



- (i) Find the x coordinate of the point A . 1
- (ii) Find the area of the shaded region bounded by the two parabolas and the x axis. 3
- (c) The population, P , of Booby Birds on Christmas island is decreasing at a rate proportional to P .
- That is $\frac{dP}{dt} = -kP$, where k is a positive integer and t is measured in years and $P = Ae^{kt}$.
- In January 2000 there were 3000 Booby Birds and by January 2010 the population had decreased to 2750.
- (i) Show that the value of k is 0.0087, correct to 4 decimal places. 2
- (ii) If the population continues to decrease at this rate what will be the expected population in 2020? 2
- (d) Use Simpson's Rule with 5 function values to approximate $\int_1^9 (\log_e x)^2 dx$, giving your answer correct to 2 significant figures. 4

Question 14 (15 marks) **Use a separate writing booklet** **Marks**

- (a) Find the values of x for which the function $y = x - x^2$ is decreasing. 2
- (b) The 8th term of an Arithmetic Progression is 23 and the 11th term is four times the 3rd term. Find the first term and the common difference. 3
- (c) After time t years the value \$ V of a car is given by $V = 24000e^{-0.1t}$.
- (i) Find the decrease in the value of the car during the fourth year. 2
- (ii) Find the percentage decrease in the value of the car during the fourth year (to 2 significant figures). 2
- (d) In $\triangle ABC$, $AC = BC$ and $\angle BCA = 90^\circ$. In $\triangle CDE$, $DC = EC$ and $\angle ECD = 90^\circ$. DA and BE intersect at F .

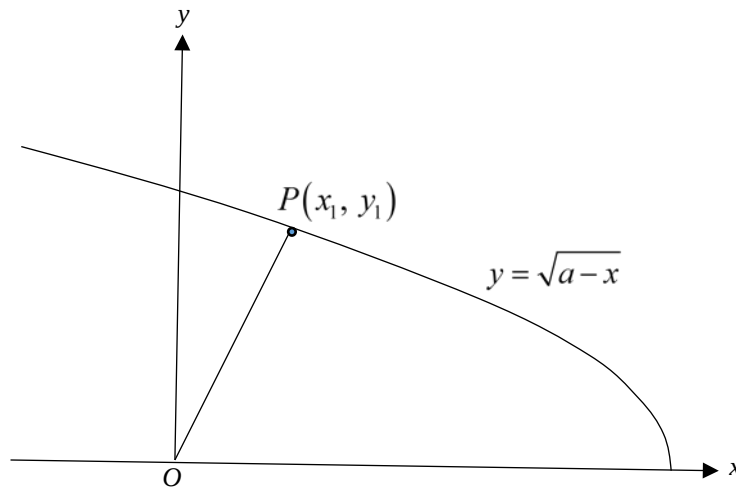


- (i) Prove that $\triangle BCE \equiv \triangle ACD$. 3
- (ii) Show that $DA \perp BE$. 3

Question 15 (15 marks)**Use a separate writing booklet****Marks**

(a) Find in simplest exact form the value of $\int_0^{\log_e 3} e^{2x} dx$. **2**

- (b) The diagram shows the graph of the curve $y = \sqrt{a-x}$ where $a > 0$. The normal to the curve at the point $P(x_1, y_1)$ passes through the origin $O(0,0)$. By considering the gradient of OP in two different ways, find the value of x_1 . **3**



(c) (i) Solve the equation $\sin x = \cos x$ for $0 \leq x \leq 2\pi$. **2**

- (ii) On the same diagram, sketch the graphs of the curves $y = \cos x$ and $y = \sin x$ for $0 \leq x \leq 2\pi$, showing clearly the intercepts on the coordinate axes. **2**

- (iii) Find in simplest exact form the area of the region enclosed by the curves $y = \sin x$ and $y = \cos x$ for $0 \leq x \leq \frac{\pi}{4}$. **3**

(d) Solve the equation $2 \log_2 x - \log_2 (x+4) = 1$. **3**

Question 16 (15 marks)**Use a separate writing booklet****Marks**

- (a) Ozzie takes out a loan of \$400 000. The loan is to be repaid in equal monthly repayments of \$3600. Reducible interest is charged at 6% p.a. calculated monthly. Let A_n be the amount owing after the n^{th} repayment.

(i) Show that $A_3 = 400\,000(1.005)^3 - 3600(1 + 1.005 + 1.005^2)$. 2

- (ii) Find correct to the nearest month the time taken for Ozzie to repay the loan. 3

- (b) A cylindrical container closed at both ends is to be made from thin sheet metal. The container is to have a radius of r cm and a height of h cm such that its volume is $2000\pi \text{ cm}^3$.

(Such a cylinder closed at both ends has surface area S and volume V given by the formulae $S = 2\pi r^2 + 2\pi rh$ and $V = \pi r^2 h$.)

(i) Show that the area of sheet metal required to make the container is 2

$$\left(2\pi r^2 + \frac{4000\pi}{r}\right) \text{ cm}^2.$$

- (ii) Hence find the minimum area of sheet metal required to make the container (leave answer in terms of π). 4

- (c) Given that the limiting sums S_1 and S_2 of the series both exist, where

$$S_1 = 1 + \sin^2 x + \sin^4 x + \sin^6 x + \dots$$

$$S_2 = 1 + \cos^2 x + \cos^4 x + \cos^6 x + \dots$$

(i) Show that $S_1 = \sec^2 x$ and $S_2 = \operatorname{cosec}^2 x$. 2

(ii) Show that $S_1 + S_2 = S_1 S_2$. 2

End of paper

2018. HSC Trial Mathematics,

M.C.

①

D

②

C $\rightarrow -1 + \frac{(-1)^2(2)^2}{-1 + 4} + \frac{(-1)^3(3)^2}{-9 + 16} + \frac{(-1)^4(4)^2}{16} = 10$

③

D

$\rightarrow$ for +ve definite $b^2 - 4ac < 0$

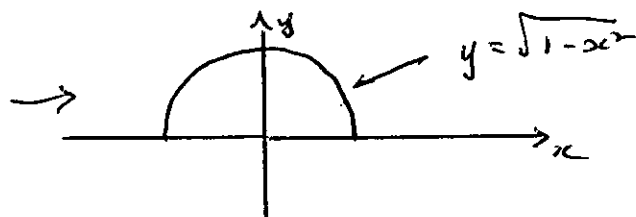
$\therefore$ (D)

④

B

⑤

B



⑥

C

$$\log x^2 - \log 3e = \log 9$$

$$\log \frac{x^2}{3e} = \log 9$$

$$x^2 = 27e$$

$$x^2 - 27e = 0$$

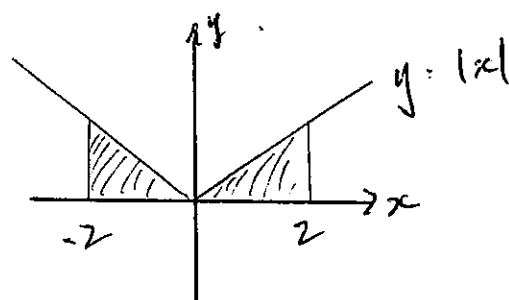
$$x(x - 27) = 0$$

$$\therefore x = 0 \text{ or } 27$$

$$x \neq 0 \therefore \underline{x = 27}$$

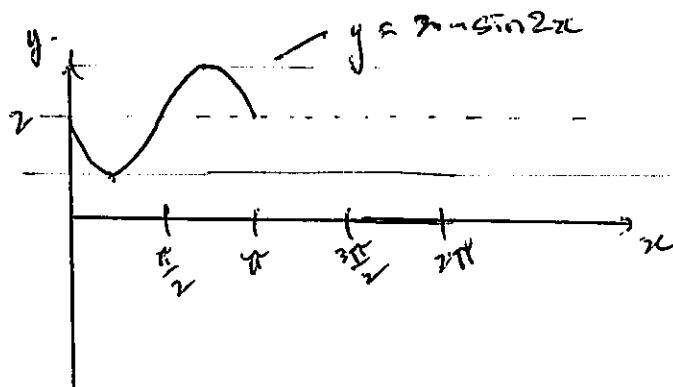
⑦

C



$$\therefore \int_{-2}^2 |x| dx = 2 \int_0^2 x dx = 4$$

⑧ A



⑨ A.

$$\frac{d}{dx} (e^{2x} \tan x)$$

$$u = e^{2x}$$

$$u' = 2e^{2x}$$

$$v = \tan x$$

$$v' = \sec^2 x$$

$$e^{2x} \sec^2 x + \tan x 2e^{2x}$$

$$e^{2x} (\sec^2 x + 2 \tan x)$$

from $\sin^2 x + \cos^2 x = 1$
 then $\tan^2 x + 1 = \sec^2 x$
 subs for $\sec^2 x$

$$e^{2x} (\tan^2 x + 2 \tan x + 1)$$

(perfect square)

$$e^{2x} (\tan x + 1)^2 = A.$$

⑩ c

$$x(x-2) \log_e x = 0$$

$$\therefore \left. \begin{array}{l} x=0 \\ x-2=0 \end{array} \right\} \begin{array}{l} x=0 \\ x=2 \end{array}$$

$$\log_e x = 0 \quad \therefore x=1$$

$x \neq 0$

$\therefore 2 \text{ roots}$

Question 11

$$\begin{aligned} \text{(a)} \quad (3 - 2\sqrt{2})^2 &= 3^2 - 2(3)(2\sqrt{2}) + (2\sqrt{2})^2 \\ &= 9 - 12\sqrt{2} + 8 \\ &= \boxed{17 - 12\sqrt{2}} \end{aligned} \quad \sim \textcircled{2}$$

$$\begin{aligned} \textcircled{b} \quad 2x^2 - 5x - 3 &= 0 \\ &\quad \left. \begin{array}{l} x - 6 \\ + -5 \end{array} \right\} \text{PSF} \\ \therefore \frac{(2x-6)(2x+1)}{2} &= 0 \quad (\text{or similar techniques}) \\ (x-3)(2x+1) &= 0 \\ \therefore x &= 3 \text{ or } -\frac{1}{2} \end{aligned} \quad \sim \textcircled{2}$$

$$\begin{aligned} \text{(c)} \quad \textcircled{i} \quad \frac{d(\sin(1-x))}{dx} &= \cos(1-x)(-1) \\ &= \boxed{-\cos(1-x)} \end{aligned} \quad \sim \textcircled{1}$$

$$\begin{aligned} \textcircled{ii} \quad \frac{d x \log_e x}{dx} &\quad \text{let } u = x \\ &\quad u' = 1 \\ &\quad v = \log_e x \\ &\quad v' = \frac{1}{x} \end{aligned}$$

$$\begin{aligned} &\frac{x}{x} + \log_e x (1) \\ &= \boxed{1 + \log_e x} \end{aligned} \quad \sim \textcircled{1}$$

Q11 (cont'd)

$$(d) \quad y = \frac{x}{x-1} \quad \begin{matrix} u \\ v \end{matrix}$$

$$\begin{aligned} u &= x \\ u' &= 1 \\ v &= x-1 \\ v' &= 1 \end{aligned}$$

$$\text{Then } y' = \frac{(x-1)(1) - (x)(1)}{(x-1)^2}$$

$$= \frac{x-1-x}{(x-1)^2}$$

$$y' = \frac{-1}{(x-1)^2}$$

$$f'(z) = \frac{-1}{i^2} = \underline{-1} \quad \therefore \underline{m = -1}$$

$$y = mx + b$$

$$b = y - mx$$

$$= 2 - (-1)(2)$$

$$= 4$$

$$\therefore \boxed{y = -x + 4}$$

~ (3)

$$\begin{aligned}
 (c) \quad \int_1^2 \frac{x+1}{x} dx &= \int_1^2 \frac{x}{x} + \frac{1}{x} dx \\
 &= \int_1^2 1 + \frac{1}{x} dx \\
 &= \int_1^2 1 dx + \int \frac{1}{x} dx \\
 &= [x]_1^2 + [\ln x]_1^2 \\
 &= (2-1) + \ln 2 - \ln 1
 \end{aligned}$$

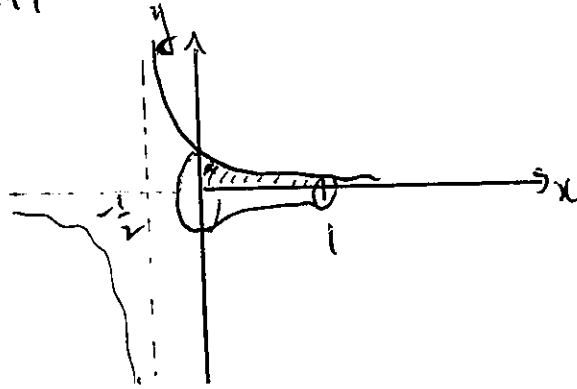
$$= 1 + \ln 2$$

3

(9)

$$y = \frac{1}{2x+1}$$

x axis, $x=0$, $x=1$,



$$V = \pi \int_a^b y^2 dx$$

$$= \pi \int_0^1 \frac{1}{(2x+1)^2} dx$$

$$= \pi \int_0^1 (2x+1)^{-2} dx$$

$$= \pi \left[\left(-\frac{1}{2}\right)(2x+1)^{-1} \right]_0^1$$

$$= -\frac{\pi}{2} \left[\frac{1}{2x+1} \right]_0^1$$

$$= -\frac{\pi}{2} \left[\frac{1}{3} - \frac{3}{3} \right]$$

$$= -\frac{\pi}{2} \left(-\frac{2}{3} \right)$$

$$= \frac{\pi}{3} \text{ units}^3$$

3

Q.12

(a) Find $\int \sec x (\cos x + \sec x) dx$.

$$= \int \frac{1}{\cos x} \left(\cos x + \frac{1}{\cos x} \right) dx$$

$$= \int \left(1 + \frac{1}{\cos^2 x} \right) dx$$

$$= \int (1 + \sec^2 x) dx$$

$$= \boxed{x + \tan x + c}$$

(b) $y = x^3 - 3x^2 + 6x$

$$y' = 3x^2 - 6x + 6$$

$$y'' = 6x - 6$$

For point of inflexion $y'' = 0$

$$6x - 6 = 0$$

$$x - 1 = 0$$

$$\therefore x = 1$$

$$y(1) = (1)^3 - 3(1)^2 + 6(1)$$

$$= 1 - 3 + 6$$

$$= 4$$

$\therefore$ pt of inflexion is $(1, 4)$.

12

(c)

$$8y = x^2 - 6x + 1$$

$$(i) 8y = (x^2 - 6x + 9) - 9 + 1$$

$$8y + 8 = (x - 3)^2$$

$$8(y + 1) = (x - 3)^2$$

$$4(2)(y + 1) = (x - 3)^2$$

$$\therefore h = 3$$

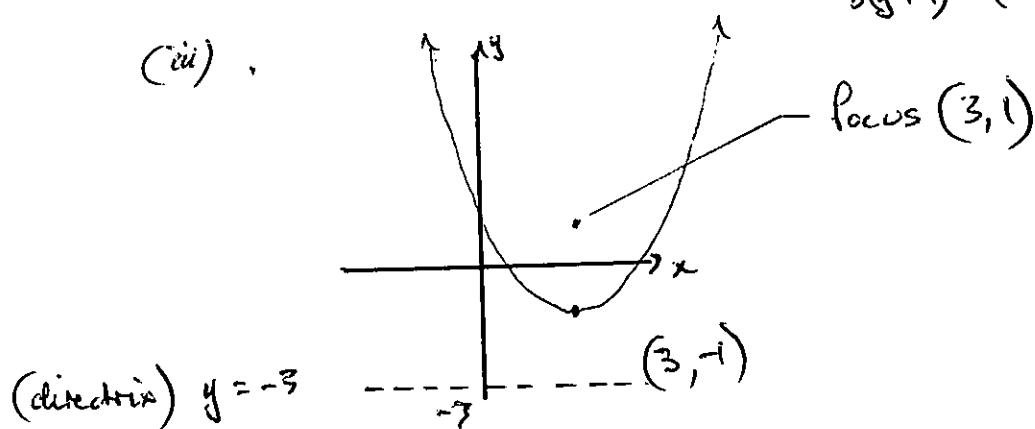
$$k = -1$$

$$(ii) \therefore \text{vertex } (3, -1)$$

$$\text{focal length } (a) = 2 \text{ units}$$

$$8(y + 1) = (x - 3)^2$$

(iii)



3

12 (d)

(iv) ABCD is parallelogram (given)

then (opposite angles of parallelogram are equal)

$$\begin{array}{lll} \text{eq BC} & 2x + y - 16 = 0 & \sim (1) \\ & y = -2x + 16 & \sim (1a) \\ \text{AB} & x - 2y - 3 = 0 & \sim (2) \\ & y = \frac{x}{2} - \frac{3}{2} & \sim (2a) \end{array}$$

~~ms~~ For $\perp$ lines

$$m_1 m_2 = -1$$

$$m_{AB} m_{BC} = \left(\frac{1}{2}\right)(-2) = -1$$

(2)

$\therefore$ perpendicular

$\therefore$ opposite angles are equal so

ABCD is a rectangle

(ii) For eqn of DC (passes through $(-1, 3)$
grad $\frac{1}{2}$, $\parallel$ to AB).

$$\therefore y = mx + b$$

$$b = y - mx$$

$$= 3 - \left(\frac{1}{2}\right)(-1)$$

$$b = 3\frac{1}{2}$$

$$\underline{y = \frac{1}{2}x + \frac{7}{2}}$$

Q12 d (cont'd)

$\therefore$ for AC intersects BC at C.

$$\therefore y_0 = \frac{1}{2}x + \frac{7}{2} \sim AC$$

$$2y = x + 7$$

$$\begin{array}{rcl} x - 2y + 7 = 0 & \textcircled{1} & \textcircled{AC} \\ 2x - 4y + 14 = 0 & \textcircled{1a} & \\ 2x + y - 16 = 0 & \textcircled{2} & \end{array}$$

$$1a - 2$$

$$-5y + 30 = 0$$

$$\underline{y = 6}$$

then subs into $\textcircled{1}$

$$x - 2(6) + 7 = 0$$

$$x = 5$$

$$\therefore C = (5, 6)$$

Q13

$y = f(x)$ is passing through $(4, 5)$

$$y' = 4x^3 + x^2 + 7$$

$$\therefore y = \int (4x^3 + x^2 + 7) dx$$

$$y = x^4 + \frac{x^3}{3} + 7x + C \quad \text{passing through } (4, 5)$$

$$5 = 4^4 + \frac{4^3}{3} + 7(4) + C$$

$$5 = 256 + \frac{64}{3} + 28 + C$$

$$\therefore C = 5 - 256 - \frac{64}{3} - 28$$

$$C = -300.\bar{3}$$

$$\therefore y = x^4 + \frac{x^3}{3} + 7x - 300.\bar{3}$$

(3)

Q.13 (b)

$$y = 4x - x^2 \quad (1)$$

$$y = x^2 - 2x \quad (2)$$

(1) - (2) ~~4~~ $0 = -2x^2 + 6x$
 $\therefore 0 = -2x(x-3)$

$$\therefore x = 0 \text{ or } 3$$

$$\boxed{\therefore x = 3}$$

(ii) . Area = $\int_2^3 (x^2 - 2x) dx + \int_3^4 (4x - x^2) dx$

$$= \left[\frac{x^3}{3} - x^2 \right]_2^3 + \left[2x^2 - \frac{x^3}{3} \right]_3^4$$

$$= \left[(9 - 9) - \left(\frac{8}{3} - 4 \right) \right] + \left[2(4)^2 - \left(\frac{4^3}{3} \right) \right] - \left[2(3)^2 - 9 \right]$$

$$= \left[0 + \frac{4}{3} \right] + \left[32 - \frac{64}{3} \right] - [9]$$

$$\boxed{= 3 \text{ units}^2}$$

13

(c) (i)

$$\frac{dP}{dt} = -kP$$

$$P = Ae^{-kt}$$

$$\text{Then } \frac{P}{A} = e^{-kt}$$

$$\ln \frac{P}{A} = -kt$$

$$\frac{\ln \frac{P}{A}}{t} = -k$$

$$\frac{\ln \left(\frac{3000}{2750} \right)}{10} = \boxed{0.0087}$$

(2)

$$\text{(ii)} \quad P = Ae^{-kt}$$

$$P = 3000 e^{-0.0087(20)}$$

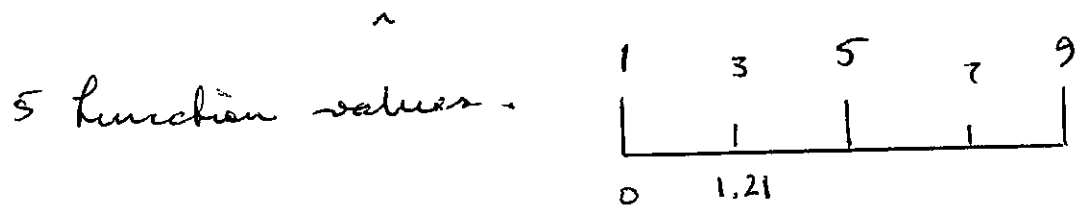
$$P = 2520.89$$

$$\boxed{P = 2520}$$

(2)

13 (d)

Simpson's Rule, $A \approx \frac{h}{3} (f_L + 4f_m + f_R)$.



$$A \approx \frac{h}{3} [1 \times f_0 + 4[f_3 + f_7] + 2[f_5] + f_9]$$

$$\approx \frac{2}{3} [1 \times 0] +$$

$$A \approx$$

$$\frac{2}{3} (29.98)$$

$$A \approx 19.99 \text{ units}^2$$

(4 s.f.)

Q 14

(a) for decreasing then $y' < 0$

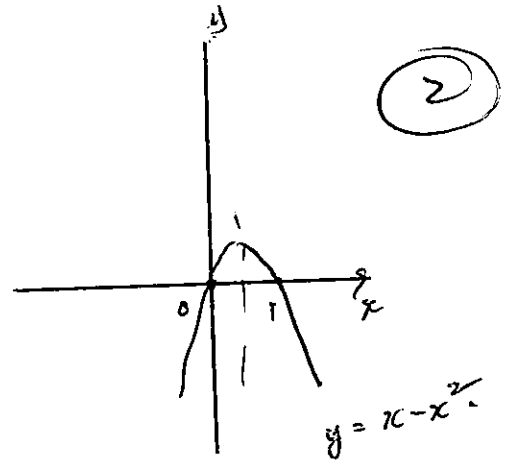
$$y = x - x^2$$

$$y' = 1 - 2x$$

$$0 > 1 - 2x$$

$$2x > 1$$

$$\boxed{x > \frac{1}{2}}$$



(b) $T_n = a + (n-1)d$

$$T_8 = 23$$

$$T_{11} = 4T_3 \quad T_{11} = T_8 + 3d$$

$$T_8 = T_3 + 5d$$

$$T_3 = T_8 - 5d$$

∴

$$\therefore 4T_3 = T_8 + 3d$$

$$4(T_8 - 5d) = T_8 + 3d$$

$$4(23) - 20d = T_8 + 3d$$

$$92 - 20d = 23 + 3d$$

$$23d = 69$$

$$\therefore \underline{d = 3}$$

14(b) (cont'd).

$$T_8 = a + n - 1 d$$

$$T_8 - (n - 1)d = a$$

$$23 - 7(3) = a = 2$$

$$\boxed{\therefore a = 2 \quad d = 3}$$

14(c)(i) $V = 24000e^{-0.1t}$

$$\therefore V_4 = 24000e^{-0.1(4)}$$

$$V_4 = \underline{16,087.68}$$

$$V_3 = 24000e^{-0.1(3)}$$

$$V_3 = 17,779.64$$

$\therefore$ depreciation in 4th year is. $17,779.64 - 16,087.68$

$$\boxed{\$1691.96}$$

$$\% \text{ dec} = \frac{1691.96}{17,779.64}$$

(in 4th year)

$$= 0.0952$$

$$\boxed{= 9.5\%}$$

(2)

14d

$$\triangle BCE \equiv \triangle ACD.$$

$$\text{let } x = \angle EEA$$

then SAS

$$\text{eg } \underline{\underline{1, (90+x), 11}}$$

$\therefore$ congruent.

let P be intersection of $\overline{AD}$ & $\overline{EC}$

$$\text{then } \angle MDC = a$$

$$\alpha \quad \angle MEF = a$$

} corresponding sides
of congruent triangles

$$\begin{aligned} \angle CMP &= \angle EMF \quad \} \text{vertically opp. angles.} \\ &= \underline{\underline{90 - a}} \end{aligned}$$

$$\text{then } \angle EFM = 90^\circ$$

interior angle sum
of a triangle

Q15

(a)

$$\int_0^{\log_e 3} e^{2x} dx$$

$$= \left[\frac{e^{2x}}{2} \right]_0^{\ln 3}$$

$$= \frac{e^{2\ln 3}}{2} - \frac{e^0}{2}$$

$$= \frac{e^{\ln 3^2}}{2} - \frac{1}{2}$$

$$= \frac{9}{2} - \frac{1}{2}$$

$$\boxed{= 4}$$

(2)

(b)

$$y = \sqrt{a-x} = (a-x)^{\frac{1}{2}}$$

$$m_{OP} = \frac{\text{rise}}{\text{run}} = \frac{y_1}{x_1} = \frac{\sqrt{a-x}}{x} \quad \sim (1)$$

$$m_{\text{tangent}} = f'(x)$$

$$f'(x) = -\frac{1}{2}(a-x)^{-\frac{1}{2}}$$

$\sim (2)$

$$\therefore (m_{OP})(m_{\text{tangent}}) = -1 \quad (\text{for perpendicular lines})$$

$$\therefore \frac{\sqrt{a-x}}{x} \times \frac{-1}{2\sqrt{a-x}} = -1$$

Q15

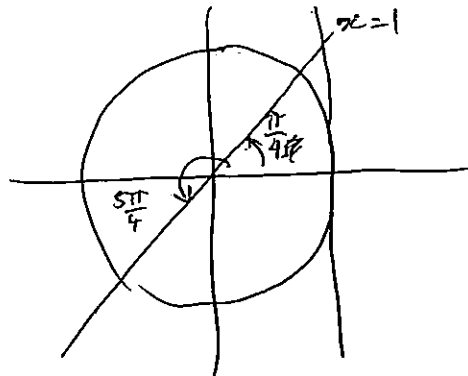
$$-\frac{1}{2x} = -1$$

$$x = \frac{1}{2}$$

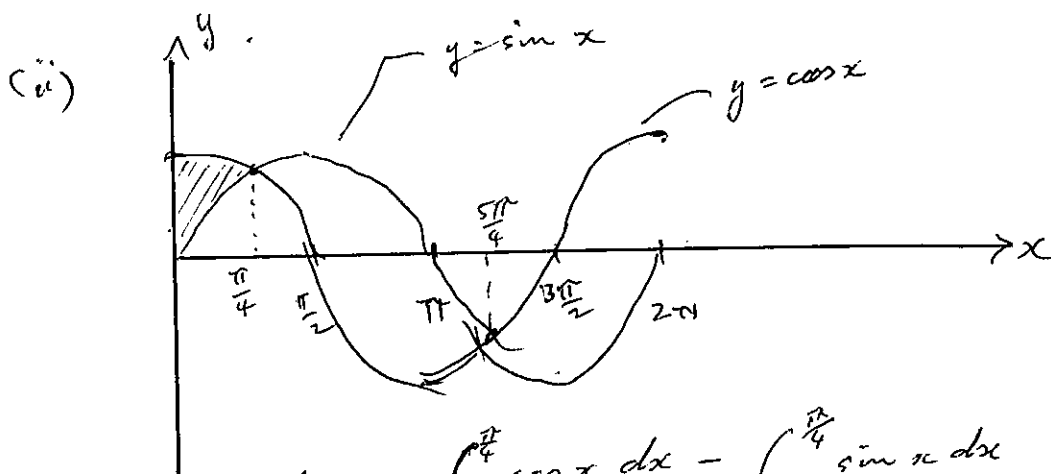
(c) (i) $\sin x = \cos x \quad \{x: 0 \leq x \leq 2\pi\}$

$\therefore \cos x \quad \tan x = 1$

Using unit circle (or otherwise)



$$\text{hence } x = \frac{\pi}{4} \text{ or } \frac{5\pi}{4}.$$



$$\begin{aligned} \text{Area} &= \int_0^{\frac{\pi}{4}} \cos x \, dx - \int_0^{\frac{\pi}{4}} \sin x \, dx \\ &= \left[+\sin x \right]_0^{\frac{\pi}{4}} - \left[-\cos x \right]_0^{\frac{\pi}{4}} \end{aligned}$$

Q15 (c) cont'd

$$\frac{1}{\sqrt{2}} - 0 - \left[-\frac{1}{\sqrt{2}} - -1 \right]$$

$$\frac{2}{\sqrt{2}} - 1$$

$$\boxed{= \sqrt{2} - 1 \quad u^2}$$

(3)

$$(d) \quad 2 \log_2 x - \log_2 (x+4) = 1$$

$$\log_2 x^2 - \log_2 (x+4) = 1$$

$$= \log_2 \left(\frac{x^2}{x+4} \right) = 1$$

$$\frac{x^2}{x+4} = 2$$

$$x^2 = 2x + 8$$

$$x^2 - 2x - 8 = 0$$

$$(x-4)(x+2) = 0$$

$$\therefore x = 4 \text{ or } -2$$

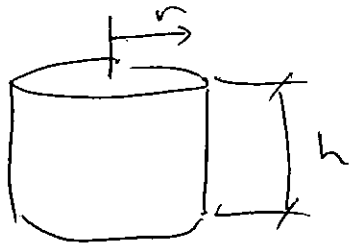
$$x \neq -2$$

$$\boxed{\therefore x = 4}$$

(3)

Q 16

⑤



$$\underline{V = 2000\pi \text{ cm}^3} \quad \sim \textcircled{1}$$

$$S = 2\pi r^2 + 2\pi r h \quad \textcircled{2}$$

$$V = \pi r^2 h \quad \textcircled{3}$$

① into ③ $2000\pi = \pi r^2 h$

$$\underline{h = \frac{2000}{r^2}} \quad \sim \textcircled{4}$$

subs ④ into ②

$$S = 2\pi r^2 + 2\pi r \frac{2000}{r^2}$$

$$\boxed{= 2\pi r^2 + \frac{4000\pi}{r}}$$

$\sim \textcircled{2}$

(i) For $A = \min$ $A' = 0$

$$\frac{dA}{dr} = 4\pi r - \frac{4000\pi}{r^2}$$

$$\frac{dA}{dr} = 0 \quad \therefore 4\pi r = \frac{4000\pi}{r^2}$$

$$r^3 = 1000$$

$$\underline{r = 10}$$

$$\text{For } r = 10 \quad S = 2\pi(100) + \frac{4000\pi}{10} = \boxed{600\pi \text{ cm}^2}$$

(c). (i)

$$S_1 = 1 + \sin^2 x + \sin^4 x + \dots$$

$$S_2 = 1 + \cos^2 x + \cos^4 x + \dots$$

$$S_n = \frac{a}{1-r}$$
$$= \frac{1}{1 - \sin^2 x}$$

$$= \frac{1}{\cos^2 x}$$
$$\boxed{= \sec^2 x}$$

$$S_n = \frac{a}{1-r}$$

$$= \frac{1}{1 - \cos^2 x}$$

$$= \frac{1}{\sin^2 x}$$
$$\boxed{= \operatorname{cosec}^2 x}$$

(2)

(2i) When $S_1 + S_2 = S_1 S_2$

$$\sec^2 x + \operatorname{cosec}^2 x = \sec^2 x \operatorname{cosec}^2 x$$

$$\frac{1}{\cos^2 x} + \frac{1}{\sin^2 x} = \frac{1}{\cos^2 x} \cdot \frac{1}{\sin^2 x}$$

$$= \frac{\sin^2 x + \cos^2 x}{\cos^2 x \sin^2 x} = \frac{1}{\cos^2 x \sin^2 x}$$

$$= \frac{1}{\cos^2 x \sin^2 x} = \frac{1}{\cos^2 x \sin^2 x}$$

$$\sin^2 x + \cos^2 x = 1$$
$$\div \sin^2 x \quad 1 + \cot^2 x = \operatorname{cosec}^2 x$$
$$\div \cos^2 x \quad \tan^2 x + 1 = \sec^2 x$$